University of Colombo School of Computing

SCS 1308 – Foundations of Algorithms

Take-Home - 03

Instructions

1. Try the following questions and upload your answer script as a zip file to the given link in the UGVLE on/before 01st November at 10 PM.
2. It's an individual task. The deadline is 10:00 PM. Submissions will be accepted until midnight with a late penalty of 5% deducted for each hour past the deadline.
3. Note: Rename your zip file as indexNo_2_last_namefname.zip.

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1. Consider the following recursive algorithm for processing a sequence of n elements.

```
Process(n)
    if n <= 1
        return
    perform 5 constant-time operations
    Process(n - 1)
```

a. Derive the recurrence equation $T(n)$, including the base case.

b. Expand the recurrence for the first four levels.

c. Solve the recurrence using the iteration method.

d. Solve the recurrence using the Substitution method.

e. Solve the recurrence using the recursion tree

f. Determine the asymptotic time complexity.

2. Consider the following recursive algorithm:

```
Process(n)
    if n <= 1
        return
    perform 5 constant-time operations
    Process(n / 2)
    Process(n / 2)
```

Answer the following:

a. Derive the recurrence equation $T(n)$, including the base case.

b. Identify the values of a, b, and $f(n)$.

c. Solve the recurrence using the Master Theorem.

d. Determine the asymptotic time complexity.

e. Explain how the 5 constant-time operations affect the final complexity.

3. Consider the recurrence:

$$T(n) = 2T(n/2) + n$$

Solve using the Master Theorem.

4. Consider:

$$T(n) = 4T(n/2) + n$$

Solve using the Master Theorem.

5. Consider:

$$T(n) = 4T(n/2) + n^2 \log n$$

Solve using the Master Theorem.

6. Consider the following recursive algorithm:

```
Search(n)
  if n <= 1
    return
  perform 3 constant-time operations
  Search(n - 2)
```

a. Derive the recurrence relation $T(n)$.

b. Solve the recurrence using the **Iteration Method**.

c. Determine the asymptotic time complexity.

7. Consider the recurrence:

$$T(n) = T(n-1) + n \qquad \text{With: } T(1) = 1$$

Use the **Substitution Method** to prove that:

$T(n)=O(n^2)$

and determine the tight asymptotic complexity.